Cross-Cultural Considerations for Designing AI Mental Health Applications: Ten Practical Recommendations

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Artificial Intelligence (AI)-powered mental health applications offer immense potential to democratize mental health services globally, especially in Low- and Middle-Income Countries (LMICs), where traditional mental health resources are scarce.[1] However, early AI applications in mental health, particularly language-based models designed to detect psychopathology, have predominantly relied on datasets sourced from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) populations, overlooking significant demographic and cultural variations (See SI).

Although cultural adaptations of digital interventions have gained attention within clinical psychology, for example, fewer than one-third of mental health chatbots explicitly define the demographic characteristics of their target users.[2] Given significant variation across populations, not only in clinical presentations but also in the availability of resources for treatment delivery, it is crucial to culturally and contextually tailor AI mental health applications to the unique culture and context of geo-cultural populations. Failure to account for these variations risks perpetuating biases and limiting the effectiveness of such tools.[3] We propose ten recommendations for developing culturally responsive AI mental health applications.

1. **Define the target population clearly:** The absence of critical demographic information such as age, gender, and race/ethnicity, in training datasets raises concerns about the generalizability of AI models across subpopulations (e.g., how performance differs across racial/ethnic subgroups[4]). Researchers should explicitly document demographic details, following established guidelines in medical and psychological research. Additionally, clear, standardized labeling practices for AI mental health applications are essential for communicating the intended user, use, and warnings about potential biases, thereby fostering user trust and informed adoption.[5]
2. **Identify cultural idioms of distress:** Building assessment or intervention applications that rely on self-disclosures about mental health skews the applicability of such applications to demographic groups with existing treatment access or high mental health literacy. This excludes disadvantaged groups that use different cultural idioms of distress (e.g., *tension*) or cultural narratives to describe their mental health problems, especially on public platforms. When developing AI-based mental health applications, collaborating with cultural psychologists, anthropologists, and local mental health practitioners is crucial to bridge this gap. Such partnerships can enable the identification, validation, and integration of culturally resonant expressions of psychopathology, ensuring these applications accurately reflect the diverse ways global communities convey psychological distress.[6]
3. **Harness multimodal data cautiously and ethically:** AI mental health applications often default to textual interactions, excluding populations that prefer or rely on alternative communication modalities—such as voice messages or video calls—including elderly individuals and people with disabilities.[7] Incorporating multimodal data sources, such as audio and video, can significantly enrich AI prediction/detection, assessment, or intervention applications by including samples from diverse socio-economic backgrounds, with the potential benefit of capturing complementary signals that are unavailable from typed messages alone (e.g., facial microexpressions, vocal tone, etc.). However, collecting and using sensitive data, via modalities like voice and video, which carry increased risks of identification and misuse, must prioritize robust privacy and

security protections through an ethics-by-design approach.[8] Moreover, clear, meaningful informed consent procedures for those contributing their data should be in place that, among other things, transparently communicate the data privacy and security safeguards and risks, such as explicitly informing participants if and how their data can be withdrawn or erased after being incorporated into model training. Finally, assessing how outcomes differ when users interact with interfaces in their native versus non-native languages can guide culturally sensitive design improvements.

4. **Engage community efforts:** Current AI mental health applications overlook the collection of culture-specific data, disproportionately impacting non-Western communities, which are often less likely to disclose mental health issues publicly due to stigma or cultural norms.[9] This disparity threatens the accuracy and applicability of language-based diagnostic models reliant on self-disclosures or public data. To improve validity, researchers should proactively engage local communities and adopt culturally sensitive data collection practices, including anonymized structured clinical interviews, widely regarded as diagnostic gold standards. Emerging AI methods, such as in-context learning, which enables models to learn effectively from small, high-quality examples, make these carefully collected datasets particularly valuable.
5. **Validate tools and check for outcome invariance**: The development of culture-specific resources could improve the validity with which relationships between language and mental health are measured across cultures. The most widely used psychosocial dictionary, Linguistic Inquiry and Word Count (LIWC), for analyzing linguistic patterns in mental health language is primarily designed for Standard American English. Similarly, current Large Language Models (LLMs) exhibit Anglocentric biases, leading to potential misinterpretations of expressed emotions.[10] Concerted efforts to culturally adapt and validate existing Natural Language Processing (NLP) tools and develop new culturally tailored resources are essential. Additionally, outcome measures employed to evaluate AI mental health applications must demonstrate measurement invariance across key demographic characteristics including gender, race, ethnicity, and religion, to ensure equitable assessments and accurate understanding of the intervention's effectiveness across diverse populations.
6. **Disentangle cultural and platform-specific effects:** Social media platform use is deeply influenced by cultural context,[11] potentially confounding AI models that rely on social media language for mental health assessment. Without distinguishing between culture-specific platform usage patterns and genuine mental health indicators, models risk inaccurate assessments and biased conclusions. Therefore, rigorous sensitivity analyses are essential—both within platforms (examining how culturally distinct linguistic features relate to mental health conditions) and across different platforms (evaluating whether mental health markers identified on one platform, like Reddit, generalize effectively to another, like Facebook). Such careful testing ensures that AI mental health assessments accurately reflect mental health conditions, rather than platform-specific or culturally mediated communication norms.
7. **Measure and enhance therapeutic alliance across cultures:** Therapeutic alliance, the quality of trust and rapport between user and mental health support, is strongly influenced by cultural and demographic factors, such as preferences for emotional expression or interaction styles. Understanding these cultural nuances can highlight crucial knowledge gaps (e.g., concerns about loss of face among Asian communities) and reveal opportunities for targeted cultural adaptations in AI mental health applications.

Recent research suggests diverse cultural preferences; for example, Chinese users may prefer emotionally expressive AI systems that foster personal connections, whereas European and American users favor more impersonal, controllable interactions.[12] LLMs can help assess therapeutic alliance[13] and detect culture-specific risks to engagement, enabling adaptive applications tailored to individual cultural contexts. However, monitoring private user conversations for alliance must also carefully balance clinical responsibilities against critical ethical concerns, notably user privacy and autonomy.

8. **Develop multicultural competency:** LLMs offer significant potential to advance multicultural competence in mental health care by highlighting cultural values and communication preferences underlying patient behaviors and treatment responses. Given patients' strong preferences for therapists sharing similar cultural or ethnic backgrounds, [14] LLMs could augment therapist training by generating culturally tailored therapeutic scenarios, identifying culturally embedded biases in clinical language, and suggesting more culturally sensitive communication alternatives, providing multiculturally competent services. For instance, AI-generated recommendations could assist therapists in adapting their interaction styles to avoid culturally specific misinterpretations, such as perceptions of rudeness arising from direct communication among traditional East Asian clients. Leveraging LLMs could enrich therapists' cultural competencies and promote more personalized, culturally attuned mental health interventions, potentially improving treatment engagement and effectiveness across diverse patient populations.
9. **Adapt to structural and contextual factors:** Effective AI mental health applications require careful consideration of structural barriers faced by real individuals, such as shared device access, limited internet connectivity, varying literacy, and language proficiency. Tailoring technologies to these contextual realities can significantly enhance user satisfaction and perceived usefulness. Establishing a centralized, transparent registry for mental health applications (similar to a leaderboard) could be a promising way to strengthen accountability and build public trust. Such a registry could disclose valuable information, including the demonstration of safety and effectiveness (e.g., clinical trial outcomes), robust privacy and security protections, and regulatory compliance (e.g., compliance with medical device regulations).
10. **Create resources beneficial to the target culture:** AI mental health research and applications must explicitly prioritize the generation of knowledge that benefits the cultures and communities they aim to serve. Researchers should contextualize mental health symptoms within specific cultural and social realities, thereby avoiding inadvertent imposition of external, often Western, frameworks, which can distort or misrepresent culturally embedded behaviors. For instance, conventional outcome measures, such as individual symptom reduction, may inadequately reflect culturally significant goals in collectivist societies, where communal harmony and family well-being are paramount. When designing AI-driven mental health applications, developers should measure outcomes consistent with the beliefs and values of the target patient populations. Furthermore, developers must rigorously adhere to applicable legal frameworks, including data privacy laws and medical device and/or AI regulations. Even when regulatory classification as wellness tools potentially bypasses stricter medical device scrutiny or a country has no comprehensive privacy law, an ethics-by-design approach remains essential, and developers can always choose to do more than what is legally required.[15] Sharing anonymized data with proper safeguards (e.g., a ban on reidentification) and trained models will help compare the generalizability of mental

health markers across cultures and translate research insights into technology.

Effective deployment of culturally equitable AI mental health applications will demand deeper interdisciplinary engagement, including ethical data governance, overcoming data silos, culturally adapted informed consent processes, robust regulatory compliance, fostering patient trust, and seamless integration into existing local mental health infrastructures. This paper aims to stimulate further interdisciplinary dialogue and research that addresses the sociotechnical complexities essential for developing and deploying equitable, culturally sensitive AI mental health applications worldwide.

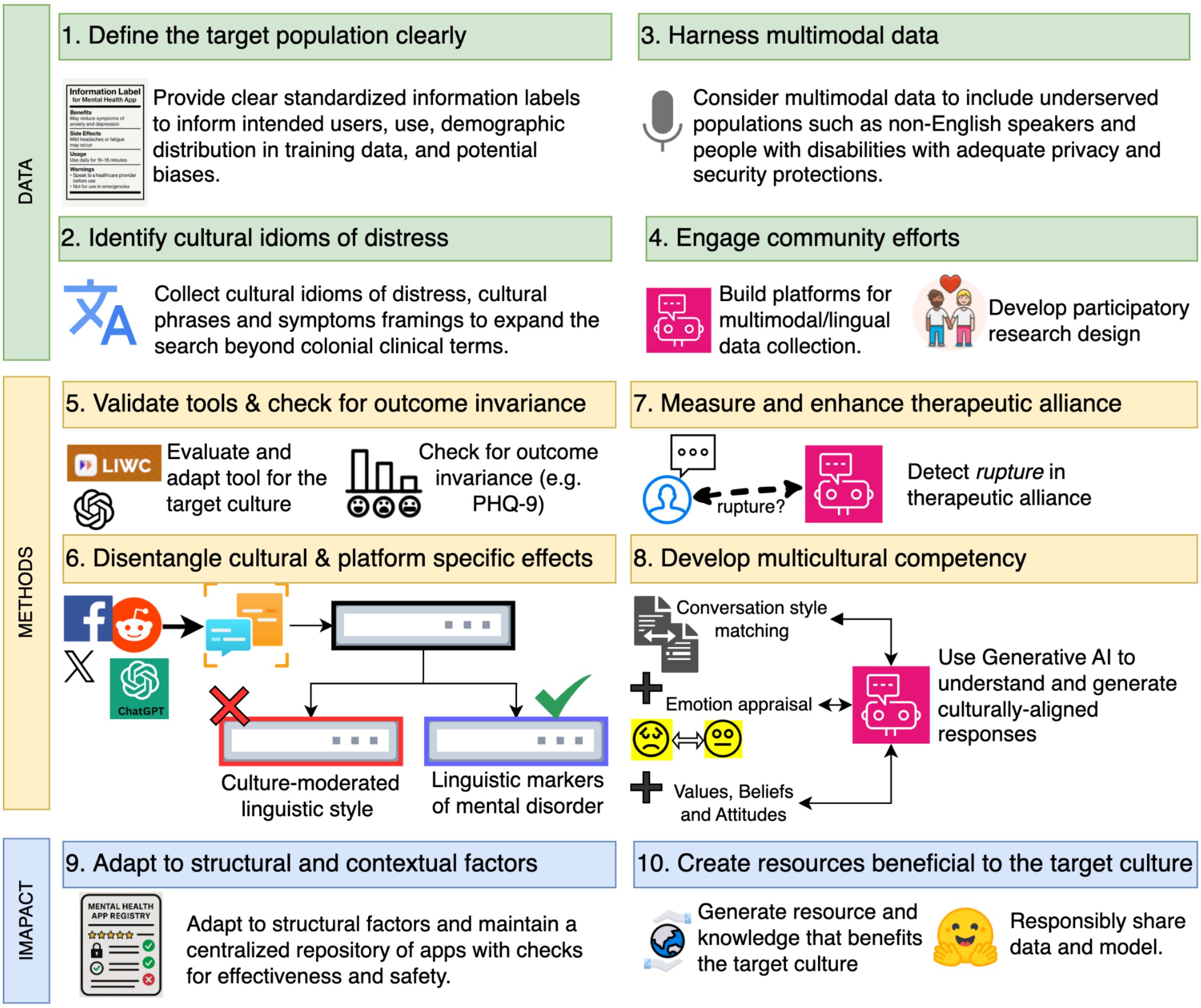

Figure 1: Ten cross-cultural recommendations for designing AI mental health applications. Recommendations are grouped by data collection (green), methodological development (yellow), and social impact evaluation (blue).

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Supplementary Information

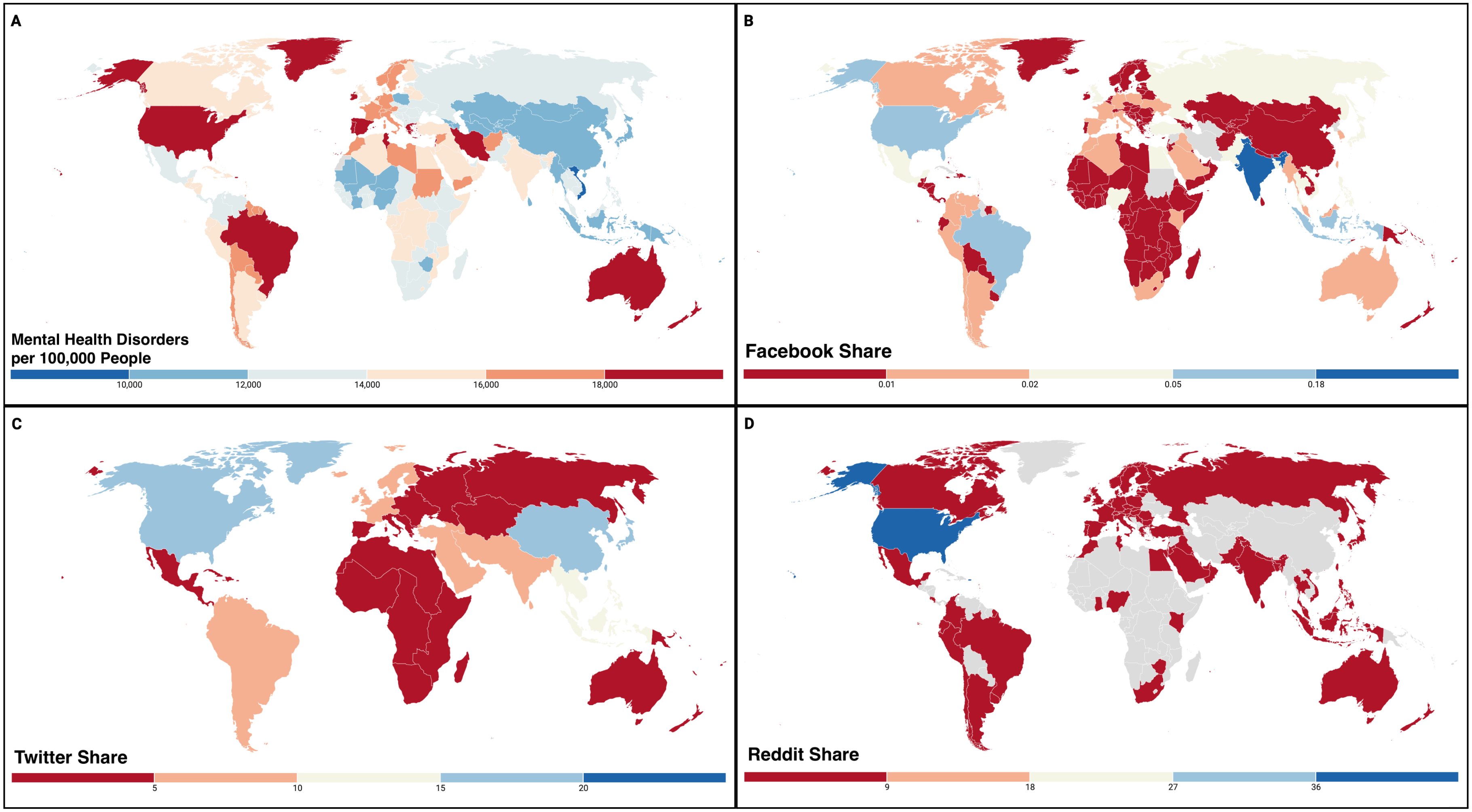

Figure S1. Prevalence of Mental Health Disorders and Social Media Platform Share Worldwide.[1] **Panel A.** Prevalence of Mental Health Disorders - Country Level, 2021. Source: (Institute for Health Metrics and Evaluation (IHME), 2015). **Panel B.** Share of Facebook Users - Country Level, July 2024. Source: (NapoleonCat, 2024) **Panel C.** Share of Twitter Advertising Audience - UN Subregion Level, April 2023. Source: (DataReportal, 2023) **Panel D.** Share of Global Reddit Traffic - Country Level, March 2023. Source: (SEM Rush, 2023). Reddit, the publicly available platform with API access, has the most traffic share from the U.S., and no traffic from China; Twitter is more widely used however, the audience share is poor for Africa, Oceania, and Eastern Europe. Facebook is most actively used in the United States, India, Southeast Asia, and Brazil.

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[1] Note that Twitter Audience Share is reported at the United Nations subregion level. As a result, subregions may include countries where platforms are banned (e.g. Twitter is banned in China but is displayed in Panel B due to its belonging in Southern Asia).